

1. Naïve follower

File: 1_naive-follower.kicad_sch

2. Tame crossover with feedback?

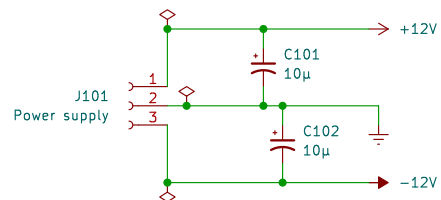
File: 2_feedback.kicad_sch

3. Feed-forward resistor

File: 3_crossover_resistor.kicad_sch

4. Class AB operation

File: 4_class_ab.kicad_sch



Transistors 101, episode 18

Kludges from Kevin's Cave

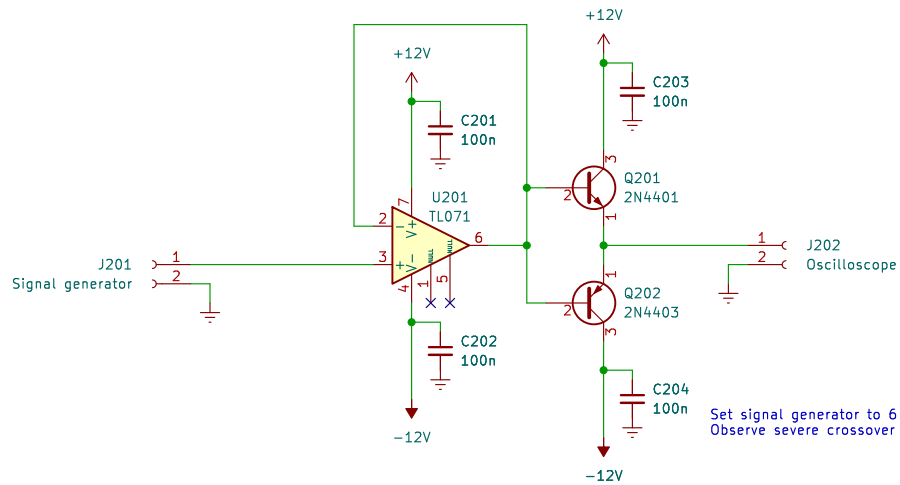
Sheet: /

File: PushPull.kicad_sch

Title: Push-Pull Followers

Size: A Date: 2026-05-13
KiCad E.D.A. kicad 7.0.11+dfsg-1build4

Rev: E
Id: 1/5



Naïve follower – just a transistor pair
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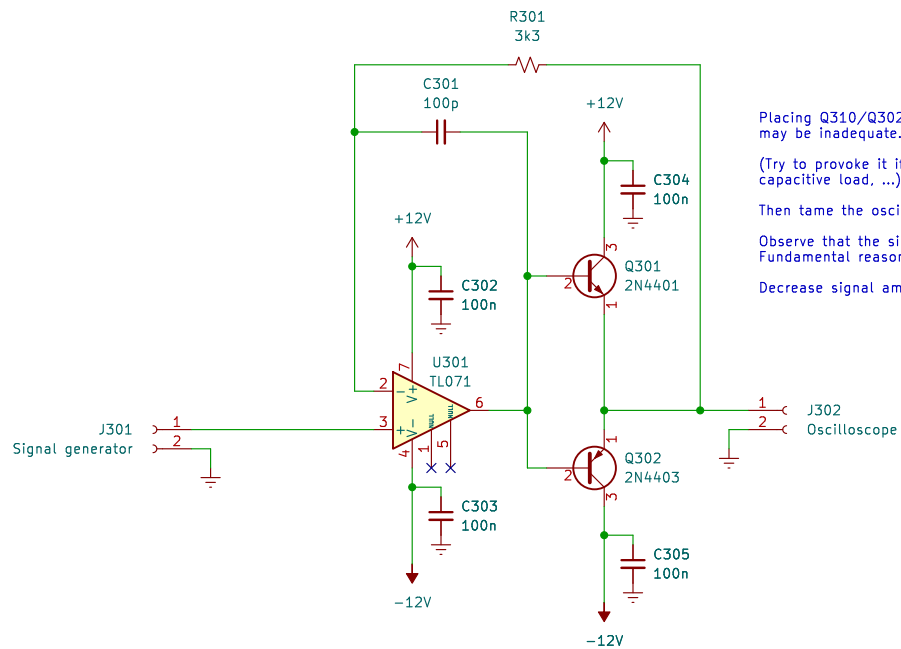
Kludges from Kevin's Cave

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Placing Q310/Q302 inside the feedback loop improves things, but compensation of U301 may be inadequate.

(Try to provoke it if necessary by pulling decoupling capacitors, adding a capacitive load, ...)

Then tame the oscillation with R301/C301.

Observe that the signal still glitches near the crossover points.
Fundamental reason: finite slew rate.

Decrease signal amplitude to 1 V_{pp} and show a mess.



Corssover distortion tamed with feedback
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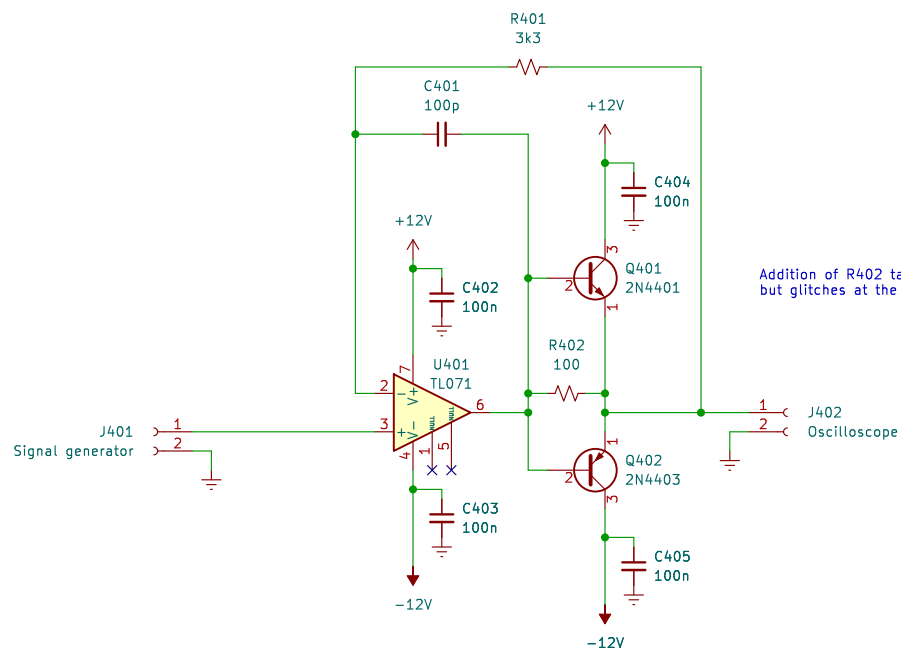
Kludges from Kevin's Cave

Sheet: /2. Tame crossover with feedback?/
File: 2_feedback.kicad_sch

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Addition of R402 tames the wild wanderings, but glitches at the crossovers remain.



Output resistor keeps output from floating near the crossover points
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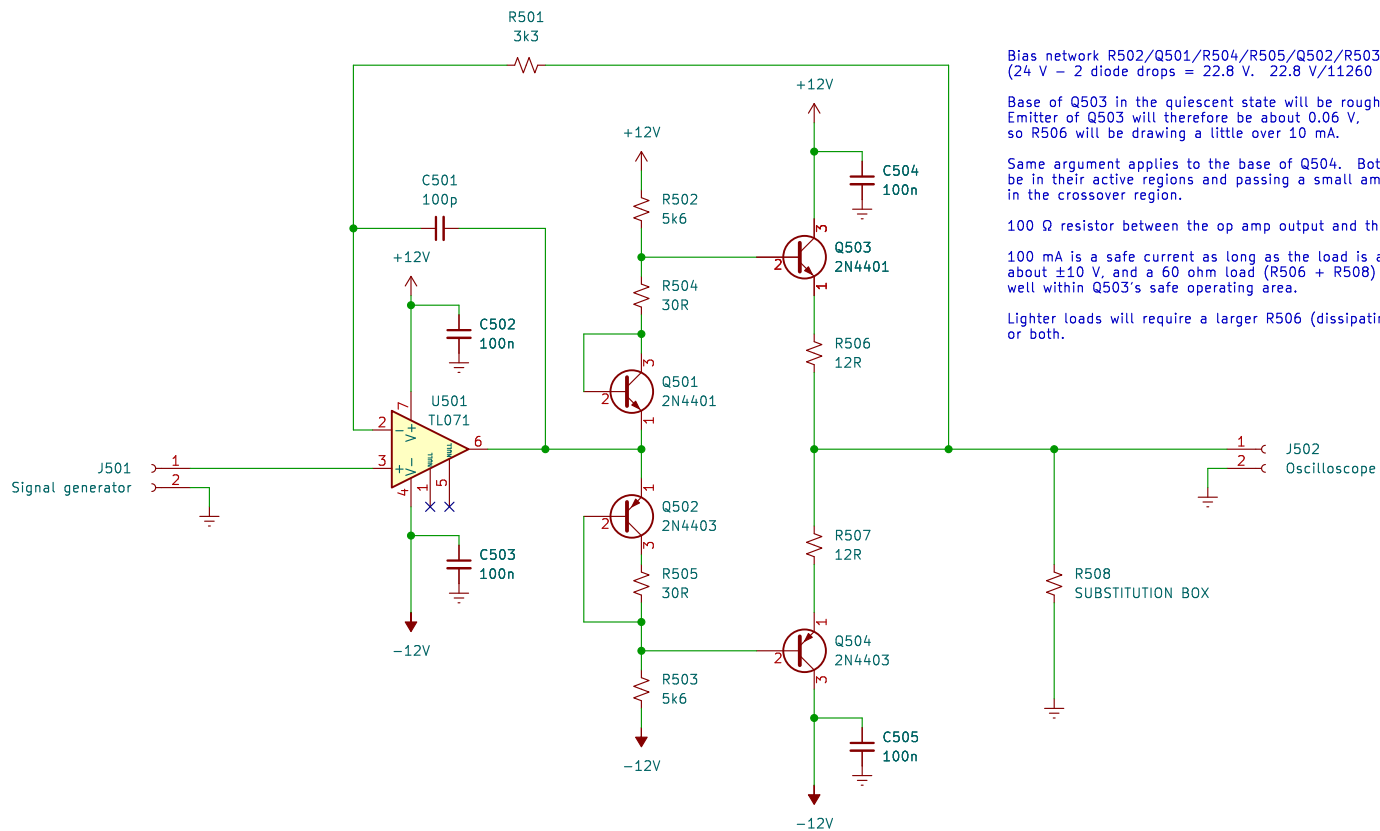
Kludges from Kevin's Cave

Sheet: /3. Feed-forward resistor/
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Bias network R502/Q501/R504/R505/Q502/R503 draws about 2 mA of quiescent current.
 $(24\text{ V} - 2\text{ diode drops} = 22.8\text{ V}, 22.8\text{ V}/11260\ \Omega = 2.0\text{ mA})$

Base of Q503 in the quiescent state will be roughly 0.66 volts, 0.06 volts above a diode drop.
 Emitter of Q503 will therefore be about 0.06 V,
 so R506 will be drawing a little over 10 mA.

Same argument applies to the base of Q504. Both transistors will therefore
 be in their active regions and passing a small amount of current when
 in the crossover region.

100 Ω resistor between the op amp output and the circuit output is no longer needed.

100 mA is a safe current as long as the load is at least 55 ohms. The op-amp clips at
 about $\pm 10\text{ V}$, and a 60 ohm load (R506 + R508) will drop 6 volts. 4 volts and 100 mA is
 well within Q503's safe operating area.

Lighter loads will require a larger R506 (dissipating power!) a restricted input voltage,
 or both.



Output transistors biased for class AB operation

Transistors 101, episode 18

Kludges from Kevin's Cave

Sheet: /4. Class AB operation/

File: 4_class_ab.kicad_sch

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Rev: E

Id: 5/5